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Using Tip of the Red Giant Branch (TRGB) Stars to measure the distance to globular clusters*

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ABSTRACT

Standard candles that lay on the instability strip allow us to probe cosmological distances via the Leavitt Law, or period-luminosity relationship. Tip of the Red Giant Branch Stars are a promising candidate which may be able to utilize a similar relationship for the various period luminosity relations that appear as stars evolve through the Red Giant Branch. We propose a new method for astronomical distances from TRGB stars using compiled literature measurements to different globular clusters and a Leavitt Law equivalent related to long secondary periods.

By querying SIMBAD and plotting RGB stars on color-magnitude diagrams, TRGB stars were identified. The ZTF database was queried for these stars and we used the analysis tool built into the database to find their pulsation periods. This allowed us to build a period luminosity relation utilizing compiled literature data on globular cluster distances. We hypothesized that a fit of the period-luminosity relationship of TRGB stars would have a p-value between 0.05 and 0.95. This would indicate that there is a period luminosity relationship for TRGB stars that can be used to calculate distance. However, none of the stars exhibited any relationship at all: the five most distinct TRGB stars all demonstrated different periods ranging between 74 and 310 days.

Keywords: TRGB, H_0, ZTF, Gaia

1. INTRODUCTION

The Hubble tension has been an issue in cosmology for the past two decades and with increased measurement certainty, the problem only worsens. The most recent measurement of H_0 from Leavitt's Law and Cepheids is $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} (Riess et al. 2022), whereas the value of H_0 from the Cosmic Microwave Background (CMB) is $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1} (Planck Collaboration et al. 2020). The increased discrepancy has motivated the search for new standard candles. One of these potential standard candles are Tip of the Red Giant Branch (TRGB) stars.

1.1. *TRGB Stars*

TRGB stars are classified as low mass ($< 2\,M_\odot$), low metallicity stars at the point before transition from the Red Giant Branch (RGB) to the horizontal branch, also known as Asymptotic Giant Brach (AGB) stars. On a color-magnitude diagram (CMD), TRGB stars lie in the redder and more luminous corner of the red giant branch, only to be superseded by red supergiants.

TRGB stars exist in a state just before they undergo a helium flash: a very brief period of intense helium fusion within their core. Due to the abrupt transition to the horizontal branch that follows the helium flash, TRGB stars are easily identifiable at the peak of the Red Giant Branch (Hence the designation, *Tip of*). The core mass of low-mass stars at the point of the He flash is well known, and is a nearly constant value, within $0.001 M_s$ (Beaton et al. 2018). Consequently, TRGB stars have nearly the same core pressure, temperature, and luminosity: the critical factor in utilizing the distance modulus to determine distances.

TRGB stars are not new to measuring distances; Harlow Shapley and Heber D. Curtis used these stars to define their distance scales for the size of the Milky Way (then considered as the size of the universe) (Madore & Freedman 2023). Since then, they have been used to measure distances to galaxies and, most recently, to measure the Hubble constant (Madore & Freedman 2023). Some of the most recent work on TRBG stars

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has been to determine H_0 by calibrating a magnitude system using V, I, J, H, and K filters resulting in an H_0 value between the one derived from the CMB and Cepheids with $H_0 = 69.08 \pm 0.8(\text{stat}) \pm 1.7(\text{sys})$ km s^{-1} Mpc^{-1} (Freedman et al. 2020).

1.2. *Standard Candles*

TRBG stars are unique standard candles because they reach a maximum luminosity of about -4 magnitudes in the I band, making them *true* standard candles compared to Cepheids which are *standardized* standard candles (Tammann et al. 2008). Since these stars do not lie on the instability strip, they may not exhibit the classical period-luminosity relationship that Cepheids and other pulsating stars do, however, RGB stars are known to pulsate like many cool red giants, just at much smaller amplitudes (Pawlak, Michał 2021). This prompts the question, is periodicity in any of these stars correlated to their luminosity? If not, would this suggest systematic uncertainty in any period-luminosity relationship?

The common method utilizing TRGB stars to calculate distance relies on the assumption that the stellar population is similar to that of the Milky Way halo for accurate measurement of distance (Salaris & Girardi 2005). We will use observations in globular clusters as a proxy for the stellar population of the Milky Way, so we can hold the assumption that this analysis on TRGB periods and luminosities holds outside of Milky Way globular clusters.

2. DATA SELECTION

2.1. *Star Queries*

Previous research identified TRGB stars by plotting their I magnitude against their $V - I$ color index on a CMD. To find suitable TRGB stars, we identified globular clusters with V and I band magnitudes in SIMBAD. These clusters were queried again for RGB stars using SIMBAD's classification catalog, providing a CMD for all RGB stars in the given globular clusters.

To choose potential TRGB stars, we performed a regional query of globular clusters with SIMBAD based on the right-ascension, declination, and angular diameter of each cluster. We limited the apparent magnitudes and declinations of the globular clusters based on ZTF's thresholds: declinations greater than -20° (visibility for ZTF) and magnitudes below 20 (limiting magnitude for ZTF) (Dekany et al. 2020). The resulting list of potential TRGB stars was queried in the ZTF DR22 objects catalog to gather time-series data for analysis of potential periods.

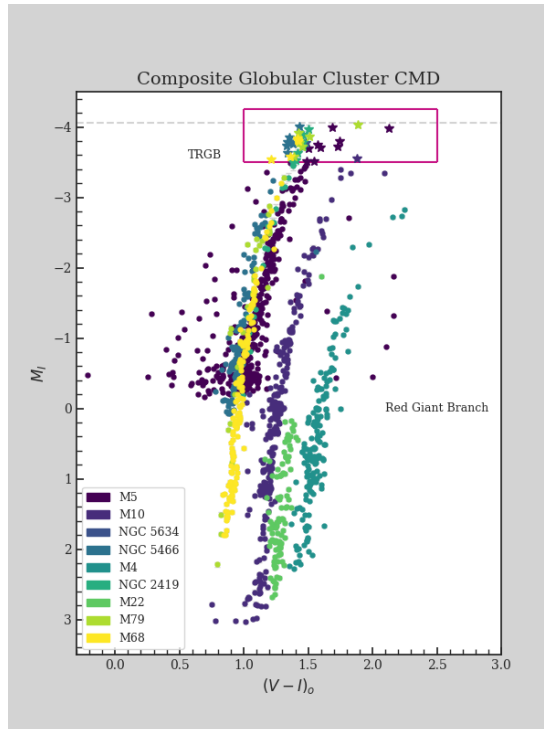

Figure 1. This plot shows the combined color-magnitude diagram for the globular clusters selected. The y-axis is absolute magnitudes whereas the x-axis is observed V-I magnitudes. The colors of the stars are sorted by globular cluster. The star-shaped points are stars that pass the TRGB criteria adapted from (Madore & Freedman 2023) which has limits of -4.5 to -3.5 I magnitudes and 1 to 2.5 color magnitudes.

2.2. *Magnitude Selection*

Our first step was to use a set of compiled data from Gaia, HST, and past literature (Baumgardt & Vasiliev 2021) to determine the distance to different globular clusters that contain observable TRGB stars. Then, the absolute magnitudes of the TRGB stars were calculated using the distance modulus and distances from previous literature (Baumgardt & Vasiliev 2021) with the following equation:

$$M = m - 5log(D/10)$$

where M is the absolute magnitude, m is the apparent magnitude, and D is the distance in parsecs. Applying this formula to all the stars in each cluster gave us absolute magnitudes in the I filter of each star. This study assumed that the TRGB stars within the same globular cluster are equidistant from Earth (Salaris &

Girardi 2005). We identified candidate TRGB stars by selecting stars with absolute magnitudes < -3.5 in the I band, which is the threshold set by previous literature (Madore & Freedman 2023).

Table 1 contains the list of TRGB stars, their periods, and their absolute magnitudes.

3. ANALYSIS

The ZTF database contains time series data, facilitating the analysis of stellar periods. Plugging the coordinates of the identified TRGB stars into the NASA/IPAC Infrared Science Archive for the ZTF DR22 Objects returns the magnitudes and time for each given star. To identify the periods, we employed a Lomb-Scargle transform which decomposes the data into a series of frequencies resulting in a periodogram where the power of a frequency tells how strong that frequency plays in determining the light curve for that object (this is essentially a Fourier transform for non-equally spaced data). The frequency with the highest power is the long-period pulsation we expect from TRGB stars. The use of the ZTF green filter for the light curve, instead of an unavailable infrared measurement, did not affect any periodicity of the star, as the star should be periodic across all wavelengths. If the star is indeed periodic, only the shape and amplitude of the light curve should change with respect to filter.

We display a phase-folded light curve and periodogram for Cl* NGC 2419 SFK 14, a star in M68, in Figure 2. To determine the legitimacy of the periods, we sorted stars by their prominence, which we defined as peak power over the average power, and established a prominence threshold of 10 to determine if a star's period is legitimate and not an alias period from data sampling or intrinsic noise. The uncertainty in our absolute magnitudes was derived by error prorogation of the SIMBAD I-magnitude uncertainty and the reported error of the compiled literature distance measurements. To determine the uncertainty in the periods, we employed a Monte Carlo approach. Our method added noise proportional to the periodogram's signal to noise ratio (SNR) to the periodogram and recorded the standard deviation of the highest power period as our uncertainty. The potential range of frequencies for the Lomb-Scargle transform are between 0.05 and 0.002 (1/days) since this range is expected for potential periods of TRGB stars (100s to 1000s of days). This range fits within the range of periods observable by ZTF, as the ZTF catalog only began observing in 2018 and a general rule of thumb for finding periods requires observations over multiple cycles limiting the maximum period ZTF can detect (Birney 1991). This means limiting our potential periods to about 500 days for ZTF to have observed for at least 4 cycles.

Plotting periods and luminosities of the TRBG stars, a linear fit returns a slope relating the star's luminosity and period. A p-value for this linear relationship is then calculated and supports whether or not a period luminosity relation exists for TRGB stars.

4. RESULTS

Only eight potential TRGB stars had an absolute magnitude below -3.5 and a prominence over 10. These eight RGB stars additionally had high-confidence periods with narrow uncertainties; with the average uncertainty of ± 4.11 days. These eight stars include one star of around 100 days, five stars between 260 and 310 days, and one star above 380 days.

Ultimately, we cannot define a comprehensive period-luminosity relationship. Only eight potential TRGB stars with magnitude less than -3.75 experience a period with a prominence greater than ten, which reveals that these stars do not exhibit sufficiently large variations in brightness for ZTF to determine or exhibit any variation at all. However, a case can be made that TRGB stars with periods between 260 and 310 days exhibit some small correlation between brightness and periodicity.

5. DISCUSSION

In our analysis of the period for the different stars, we added noise, based on the signal noise of the periodogram, to the periodogram. Then we recorded the standard deviation in the period of the highest power over 1000 trials to determine the uncertainty in the period. This method makes the assumption that there is a single dominant period and records the uncertainty due to the prominence of the peak in the periodogram. For stars that exhibit poor periodic fits, this method of uncertainty is valid as it shows large deviations in the period of largest power. However, in the case that the period is multi-modal, or has multiple periods with high powers, the method of adding noise to the periodogram may result in extremely high uncertainty, particularly if the power value is similar between the two periods as the most dominant period will be highly dependent on the

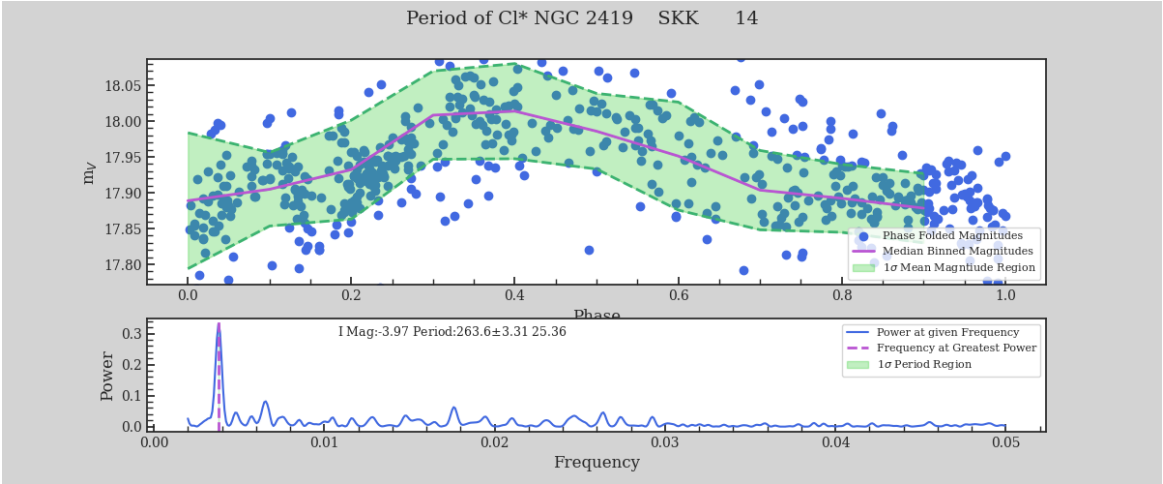

Figure 2. This plot shows the phase folded light curve of an identified TRGB alongside the periodogram. The blue plot are the magnitude measurements at the phase of best period. The purple sinusoidal curve is the sine curve for the fitted period. The left plot shows the frequency as a function of power for the data, with the best fit frequency highlighted with a purple line.

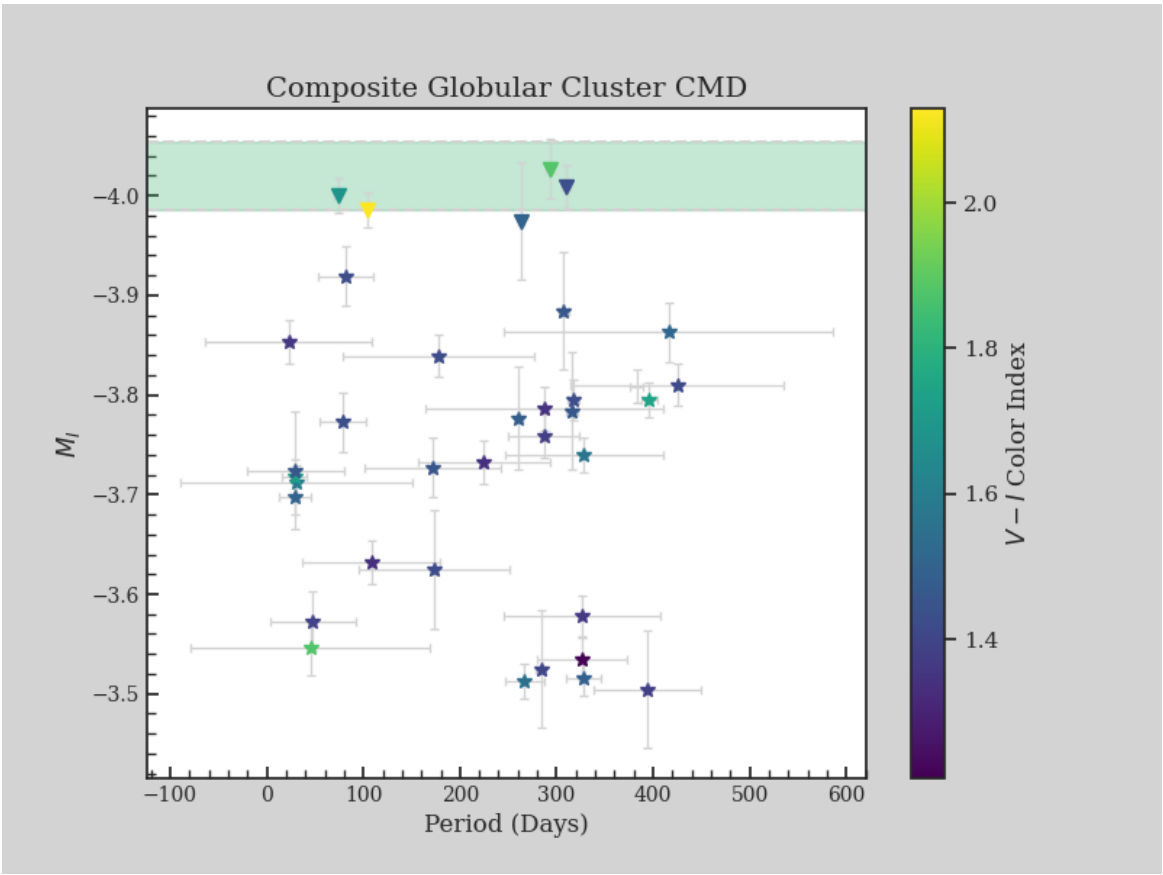

Figure 3. This plot shows the period/luminosity relationship of the 36 RGB stars within the -4.5 to -3.5 I magnitude range. The y-axis is absolute I magnitude and the x-axis is the period in days. The color of the data point corresponds to the star's $V-I$ color index. The shaded area represents the exceptionally ideal TRGB absolute magnitude region outlined by (Freedman 2021).

random noise applied close to the periods values. For analysis of multi-modal periods, a new method of determining the uncertainty in the various periods or the significance of each period will need to be used.

Table 1. Periods of TRGB stars

Cluster	Star	M_I	Period (days)	Prominence
M5	NGC 5904 SB 2-2	-3.99	104.1 ±0.55	37.02
	NGC 5904 SB 2-3	-3.81	382.9 ±6.87	19.74
	Cl* NGC 5904 SB 2-13	-3.51	266.8 ±19.1	10.83
	2MASS J15182887+0205442	-3.72	28.4 ±16.43	8.95
	2MASS J15182938+0208275	-3.52	328.0 ±15.42	8.74
	NGC 5904 865	-3.74	328.0 ±83.96	6.75
	NGC 5904 SB 2-1	-4.0	73.9 ±3.49	6.72
	Cl* NGC 5904 SAW V179	-3.80	396.1 ±7.66	6.68
	Cl* NGC 5904 SB 2-10	-3.70	29.4 ±14.24	5.48
	NGC 5904 351	-3.71	30.0 ±121.25	4.87
M10	NGC 6254 1002	-3.55	45.3 ±117.25	5.56
NGC 5634	Cl* NGC 5634 BFI 2	-3.78	260.6 ±3.61	18.83
NGC 5466	ROTSE1 J140613.69+283013.6	-4.01	310.2 ±4.61	13.45
	Cl* NGC 5466 FSBRGB 4	-3.76	286.9 ±35.79	5.08
	Cl* NGC 5466 FSBRGB 2	-3.85	22.5 ±86.68	4.37
	V* BK Boo	-3.63	108.0 ±70.45	4.51
	Cl* NGC 5466 FSBRGB 5	-3.73	224.8 ±68.99	3.83
	Cl* NGC 5466 FSBRGB 3	-3.79	286.9 ±121.6	3.69
NGC 2419	Cl* NGC 2419 SKK 14	-3.97	263.6 ±3.30	25.36
	2MASS J07380598+3852055	-3.88	307.2 ±5.22	15.96
	KGS2008 J073813.5+385257	-3.52	283.8 ±5.22	9.88
	Cl* NGC 2419 RH 28	-3.78	315.9 ±5.76	9.66
	2MASS J07381690+3853351	-3.50	394.0 ±60.46	7.37
	KGS2008 J073809.4+385210	-3.72	29.6 ±51.64	6.29
	KGS2008 J073809.6+385338	-3.62	173.5 ±78.4	5.27
M79	NGC 1904 153	-4.03	292.8 ±4.61	15.75
	Cl* NGC 1904 SAW V24	-3.73	171.1 ±71.55	9.75
	Cl* NGC 1904 SAW V2 4	-3.92	81.9 ±28.38	7.01
	KGS2008 J052408.5-243151	-3.77	78.6 ±25.3	5.46
	NGC 1904 300	-3.57	47.8 ±44.07	4.63
	NGC 1904 223	-3.86	415.9 ±168.72	4.54
M68	Cl* NGC 4590 WALK 89	-3.80	317.3 ±5.1	4.52
	Cl* NGC 4590 ALC 53	-3.54	326.5 ±45.72	3.32
	Cl* NGC 4590 HAR 1260	-3.81	424.8 ±110.11	3.18
	Cl* NGC 4590 ALC 62	-3.84	177.6 ±98.88	2.99
	Cl* NGC 4590 ALC 97	-3.58	326.5 ±82.58	2.79

Note—Table contains list of TRGB candidates from data selection, sorted by cluster and then by magnitude in I band. The 5 columns include the cluster names, star names, magnitude in I band, most likely period in days and the prominence of that period.

Additionally, Cl* NGC 5904 SB 2-2 exhibited an unusually robust period compared to the other RGB stars. The 1σ binned magnitude regions for this star are extremely tight, meaning data points strongly track the period. Furthermore, confidence in the period is very high, with a value of 104.1 ± 0.53 days, which is nearly an order of magnitude higher confidence compared to any other period. Not only is the confidence high, but the change in magnitude across the period is 1.2 magnitudes, which is also nearly a magnitude greater than any other star. This may suggest a different mechanism behind the variability, such as a binary system. Approximately 0.5% of the stars in the RGB are suspected to be contact binaries, which provides a reason for the skepticism behind Cl* NGC 5904 Sb 2-2 being included in the analysis (Wood et al. 1999).

Cepheids, Delta Scutis, and other stars that are present on the instability strip rely on either the kappa mechanism or helium shell reignition for changes in their luminosities. TRGB stars do not have a well-understood mechanism for their secondary pulsations (Pawlak, Michał 2021). Pawlak hypothesizes that late stage RGB stars, a classification which encompasses TRGB stars, exhibit a change of dominant pulsation modes at three different regions in a period-luminosity diagram. More importantly, when plotted on a CMD, TRGB stars are in a region that contains a higher density of stars experiencing LSPs (Pawlak, Michał 2021). Previous literature suggests that LSPs are primarily found as RGB and AGB stars change between the different P-L relationships they can experience as they evolve (Pawlak, Michał 2021). This may explain why our hypothesis has been rejected as the presence of LSPs usually means that the P-L relation is transitioning. Verifying the presence of long secondary periods (LSPs) in the TRGB stars we observed could serve to back this up and would be an interesting topic of further study.

There is still a lot of research to be done on the causes of TRGB star periodicity separate from finding a period luminosity relationship. In addition to this, the classification of RGB and AGB stars into different groups based on periodicity is still unclear. This actually aligns with some other previous studies using data collected by MACHO which found no period luminosity relationship in RGB and AGB stars, though a P-L relationship for RGB and AGB stars has been suggested by other sources(Derekas et al. 2006).

If results from future investigations suggest a P-L relationship, a discussion of whether these results aligns with the Cepheid period-luminosity relationship and how future work for distance TRGB stars with the James Webb Space Telescope (JWST) can confirm discrepancies between the two methods of determining distances (Baumgardt & Vasiliev 2021).

6. CONCLUSION

This study ultimately determined that TRGB stars do not have a period-luminosity relation in general. There is still a possibility that certain subgroups of stars experience periods that could be utilized for distance measurement, but this would require further research into how these stars could be classified. Future research into possible groupings for TRGB stars of different periods, processes behind LSPs and their correlation with the TRGB, and different methods for determining the uncertainty of these periods could serve as a valuable tool in understanding these standard candles and this stage of stellar evolution.

With no apparent period-luminosity relationship for TRGB stars, this warrants caution to using other period-luminosity relationships to determine distances.

Facilities: ZTF

Software: astropy

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APPENDIX

A. CODE

Access to this paper's code is provided by the following link:
https://drive.google.com/file/d/1NGuJ9lB9WVPd5p0vnMxPTYxaG9nUmj_4/view?usp=sharing
Other periodograms can be found in this notebook.